HOPPER PORTFOLIO

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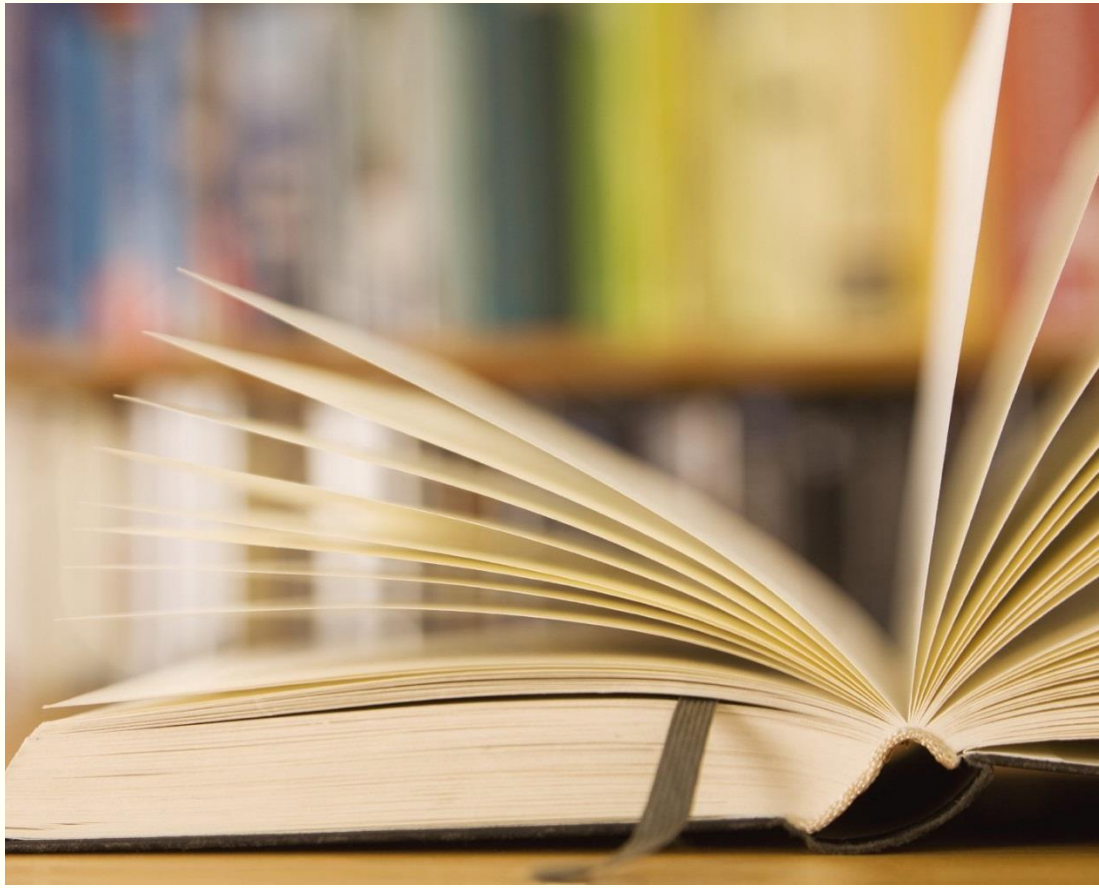
Spring 2023

Intro to Mechanical Design & Engineering (IDEA 170)



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Assignment

The hopper project tasks student with the goal of creating a fully functional machine able to jump on its own between a time frame of 8 and 60 seconds. Both the mechanism for jumping and for the timer are to be made by the student using a specified list of items. In addition, the student was tasked with incorporating a design goal to their project.

Additional constraints were to try to incorporate biological influences in our mechanism and to only use the specificed materials in our final design. Those items are listed on the following slide.

Materials

- In addition to the list of objects, students were given the option to use a rocket ignitor and small 3D printed pieces, as well as unlimited access to MDF and adhesives.

Spring 2023 Hopper Kit Items

The items below provide the maximum quantities/sizes of materials that may be used for the construction of your final hopper design. You may replenish your supplies as needed during your iterative design process.

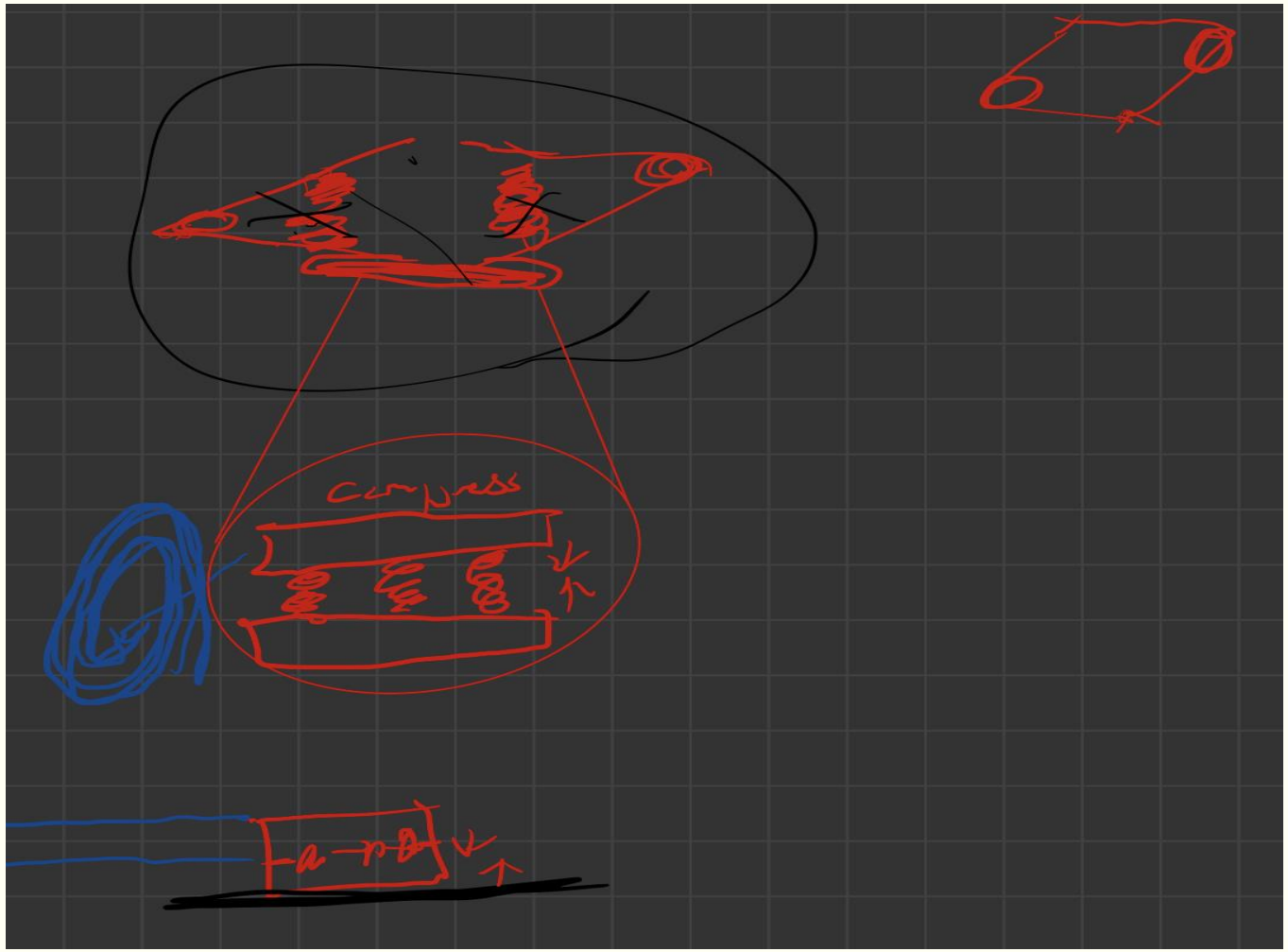
Item	Name	Quantity	Description
1	Safety glasses	1	Wear them while you work!
2	Aluminum foil	1	6in x 6in square of heavy duty aluminum foil
3	Balloon (large)	1	12in diameter
4	Balloon (long)	1	long, twisty balloon
5	Balloon (small)	2	5in diameter
6	Cable tie	4 total	black Nylon cable tie with lengths of 4" or 8"
7	Compression Spring	2	3.0" long, 5/16" outer diameter, stainless steel
8	Cotton	1	10" x 10" fabric square
9	Ear plug	1	orange noise-reduction foam ear plug
10	Fence wire	1	6in length of 14ga multipurpose
11	Fishing line	1	3ft length, 0.022in diameter, 30lb load limit, clear monofilament nylon line
12	Golf ball	1	yellow recycled golf ball
13	Magnet	2	neodymium cylindrical ¼" outer diameter x ⅛" height
14	Pencil	1	#2 wooden pencil with eraser
15	Plastic bag	1	Low density polyethylene (LDPE) foldover "sandwich sized" bag
16	Resistance Band	1	1ft length 2.4lb (tan, contains latex) or 4.7lb (green, latex free)
17	Rubber tubing	1	1 ft length of 1/4in outer diameter by 1/8in inner diameter latex rubber tubing
18	Ruler	1	6in lasercut IDEAS ruler
19	Steel ball bearing	2, total	stainless steel ball bearing with diameters of 3/8" and 1/2"
20	Torsion spring	2, total	90° torsion spring, left hand or right hand
21	Wooden dowel (small diameter)	5, total	12in long Birch dowel rod with diameters of 1/8", 3/16", 1/4"
22	Wooden dowel (large diameter)	1	6in long Birch dowel rod with a diameter of 3/4"

ITEMS BELOW ARE NOT PHYSICALLY IN KIT, BUT ARE AVAILABLE FOR USE

23	Fiberboard sheet	-	1/8in or 1/4in thick, smooth Medium Density Fiberboard (MDF) (sample in kit)
24	Super capacitor†	2	10F, 2.7V, 34mOhm equivalent series resistance, super capacitor
25	Rocket igniter†	1	Estes solar model rocket starters, or custom resistance wire
26	Water	1	100ml, tap water
27	Solder	-	0.031in diameter lead free solder with flux for joint making as needed
28	Hot glue	-	Gorilla hot glue sticks
29	Wood glue	-	Titebond III water-proof wood glue for joint making as needed
30	Epoxy	-	two-part epoxy, 5 minute
31	3D printed part‡	<3 parts, 6in^3	No more than 3 printed parts, total volume on SolidWorks < 6 cubic inches (sample in kit)
32	Thread	-	General purpose black and white thread
33	Heat shrink tubing	-	.09in inner diameter black heat shrink tubing
34	Machine screw	10, total	4-40 thread pan head phillips machine screw with lengths (in) of ½, ¾, 1.25
35	Nuts (Hex or locknut)	10, total	4-40 thread nut or nylon-insert locknut
36	Electrical wire	1	1ft length of insulated solid 22AWG hook-up wire
37	Wood Glue	-	Adhesive
38	Two-part Epoxy	-	Adhesive - mix epoxy with hardener in 1:1 ratio
39	Hot Glue	-	Adhesive

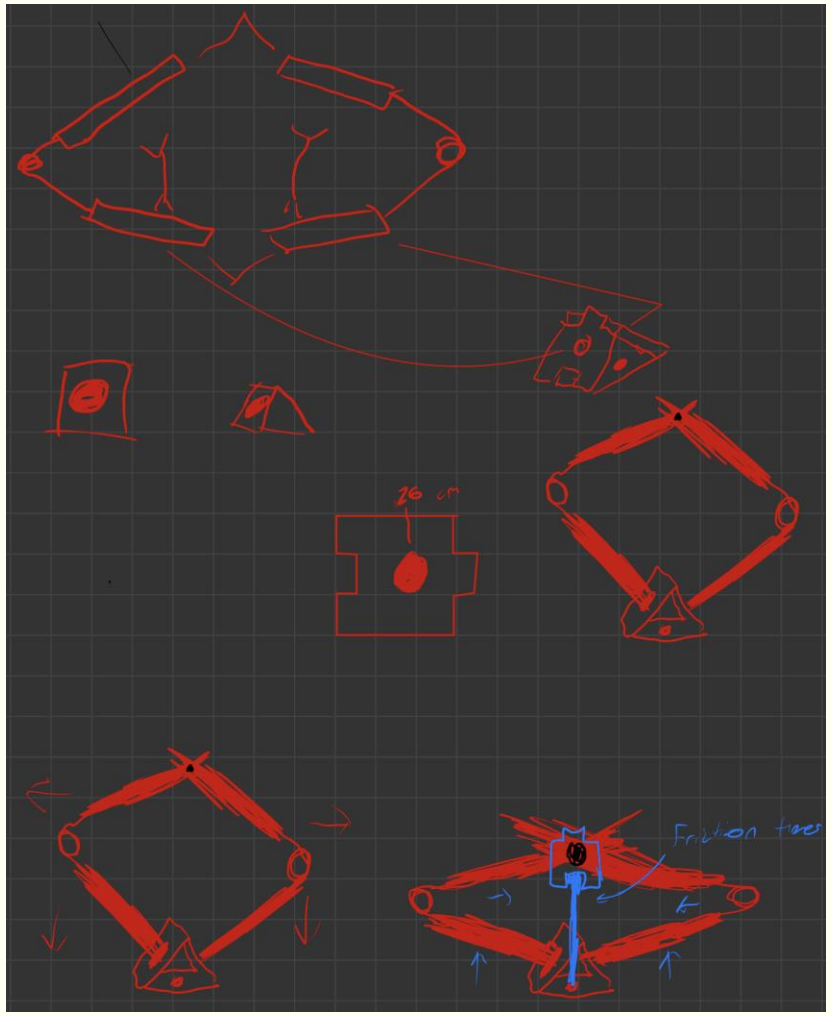
Early Ideas

I began my design process by testing the torsion springs as I felt that they could bother double as a hinge and a spring for my design. I figured that by doing this I would have a good starting point regarding how the hopping mechanism would work. The design I created resembles a grip trainer. my idea was to then try and make a sort of base to create a trampoline effect.



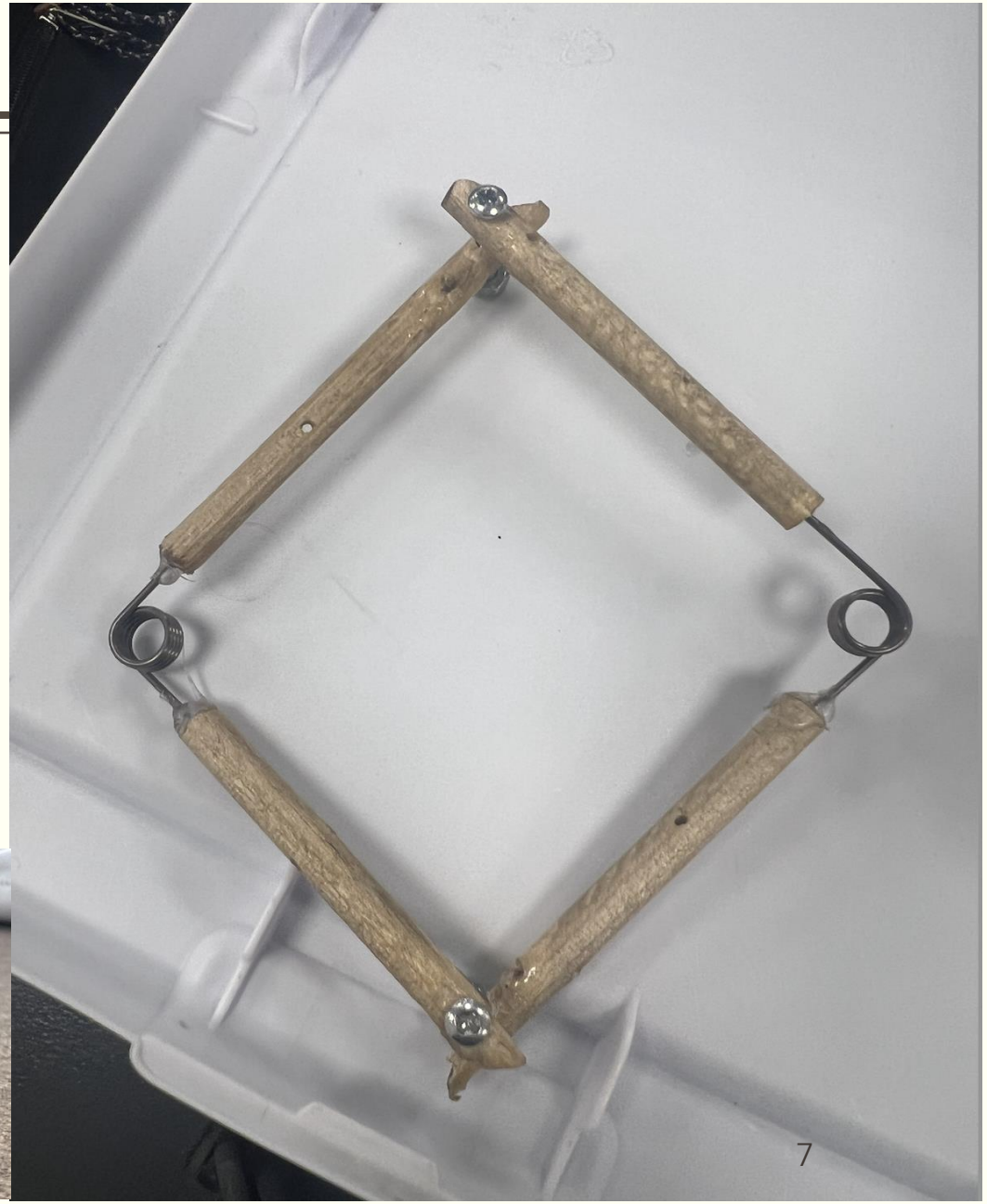
Initial Experiments

I continued to experiment with the torsion springs and attempted to incorporate a compression spring to increase the tension. However, this backfired and I decided to remove the compression springs for now. I planned to somehow attach the two hinge like pieces together using a small base using a design I made for the laser cutter



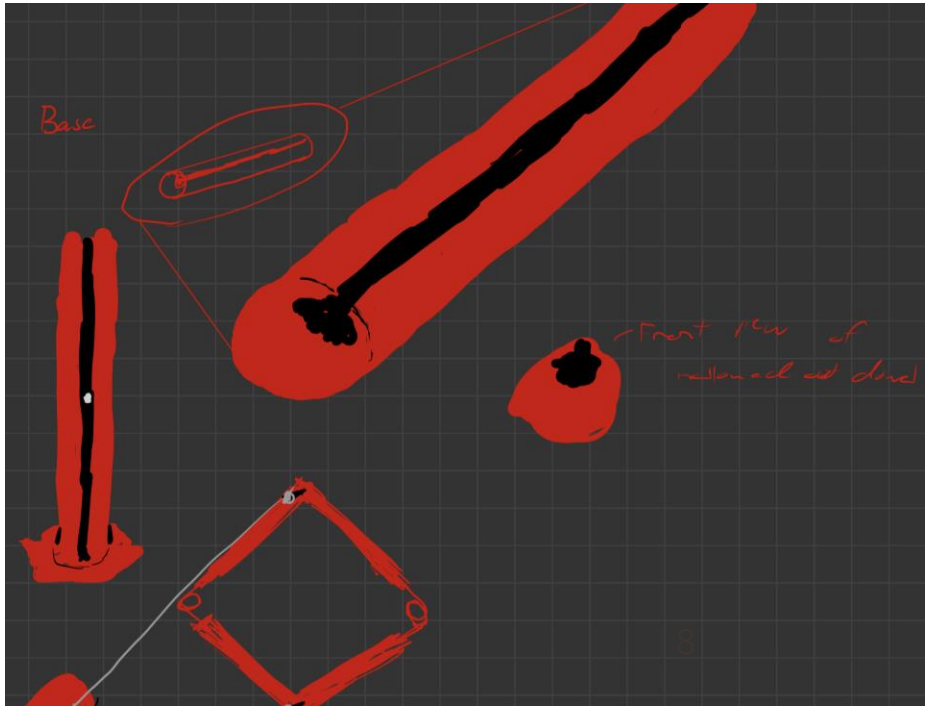
Prototype I

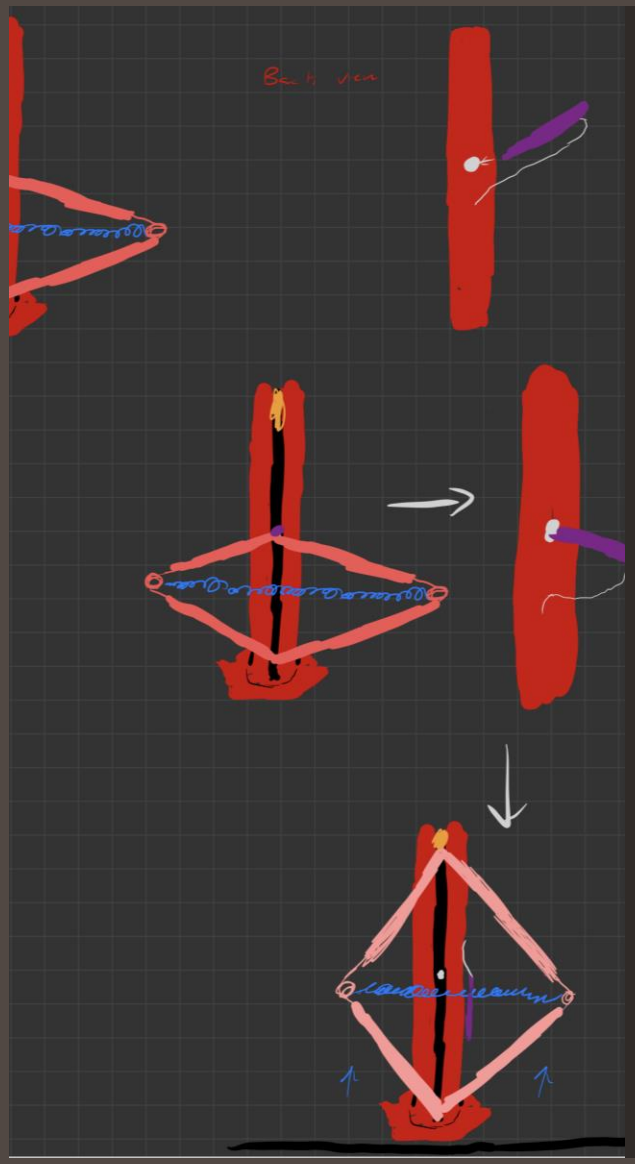
My first prototype saw the completion of my hopping mechanism. Pushing down on the top and quickly releasing saw a fast burst of energy propelling the structure upward. I next sought to deal with the issue of a timer and a way to keep the flimsy build standing. To keep the build standing I created a lego-like piece with holes in the middle to fit the dowels of my hopper in. Using a base like that helped to stabilize the structure but I was unable to create a stable model with enough give for the dowels to be able to move. This in tandem with the looming issue of my timer led to me trying a different approach.



Prototype II

- In my second major rendition of my hopper, I incorporated a stronger and larger base to keep it stable. The based was made using a rather thick dowel and a small piece of MDF. I used the drill press as well as a small file to create a hole that I carved into a sort of threads for the screws on the hopping.



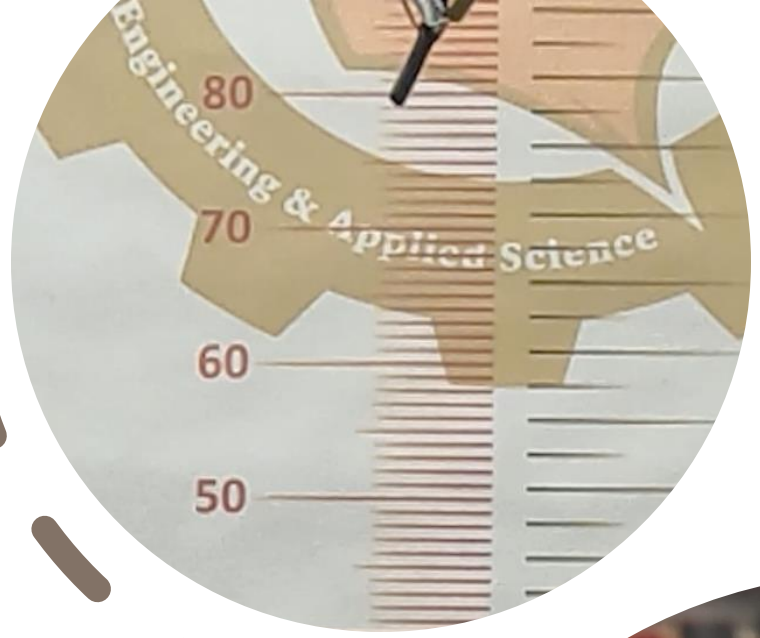


Further Ideas and Pitfalls

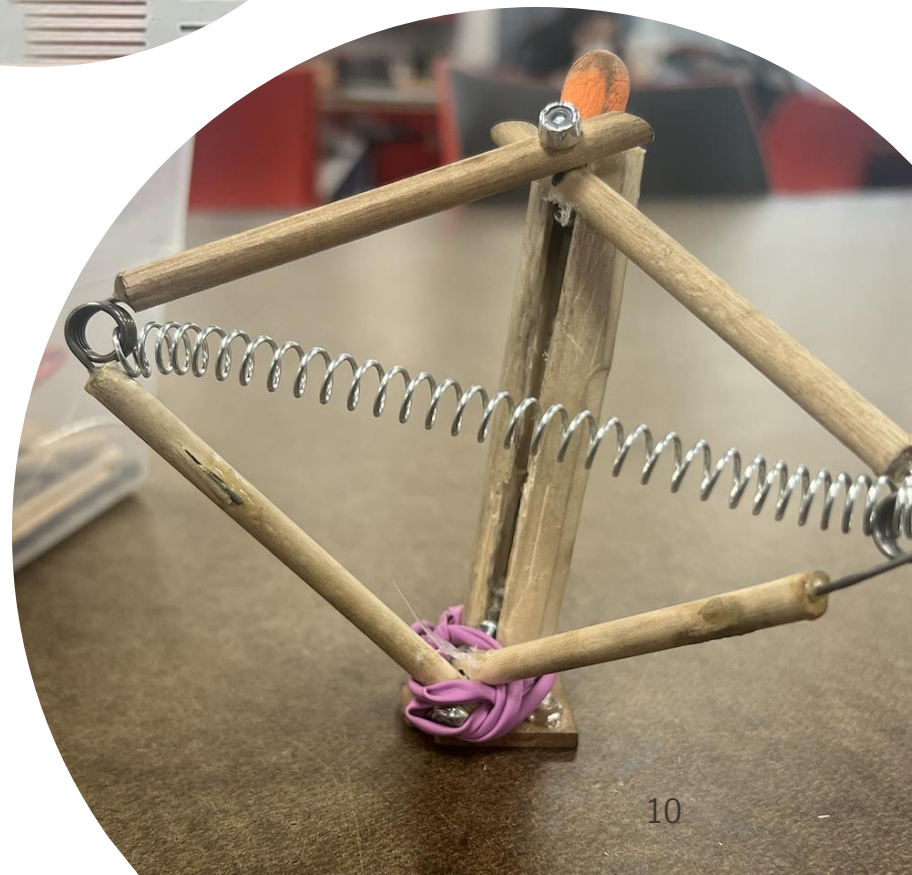
- The greatest struggle of this hopper is its inaptitude to most timers. Most of my attempts at a friction timers failed as the strength of the torsion springs were far greater than that of the timers I had attempted. This problem did however lead me to a method of increasing the tension of my hopper. By incorporating a compression spring attached to either torsion spring cause the stored energy to be rocketed upwards with much greater acceleration than earlier renditions. Since jump high was a personal goal for me outside of the project, I was overjoyed with this discovery, but it proved to only increase my struggle to find a functioning timer.

Final Product

- My final hopper design followed the same structure as the previously shown version did with some small changes. The compression spring being stretched, the ear plug used to cap the top of it and the balloon used to further attach the spring part to the base of the hopper. I unfortunately was unable to find a consistent timing method for my hopper. The hopper did jump rather high, and I was able to stay mostly true to my design goal of using primarily material found within our starter boxes. I reach a jump high of around 60 cm (slightly skewed by the camera angle).



The jump achieved is primarily due to the incorporation of the compression spring. This caused the entire mechanism to accelerate upward comparatively more than the earlier iterations. One unintended side effect of this was the fact that this tension often damaged later renditions cause me to remake it more times than any other version I had made.



Energy Calculations.

Ustored= mgh

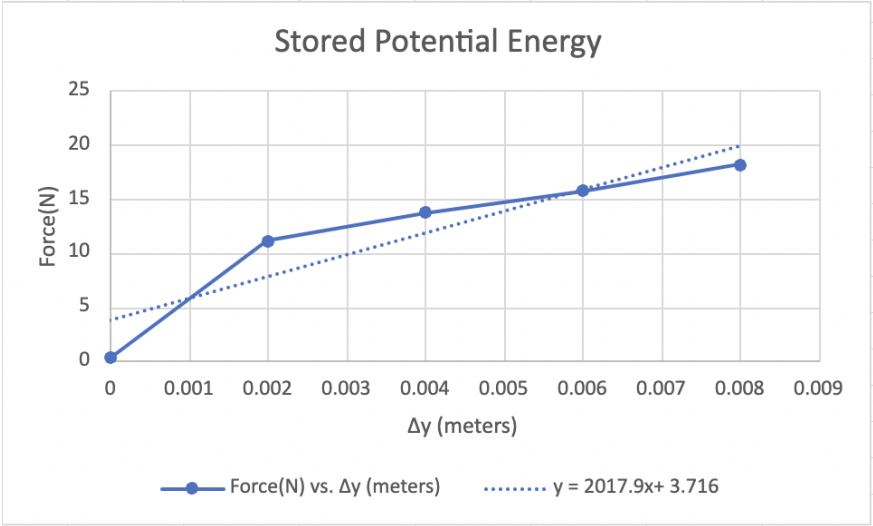
0.102 N m=.028*h

Theoritcal height = 37 cm

Actual jump height = 60 cm

Efficiency = 162.1%?!

This impossibly large efficency of this hopper is that the methods used of recording energy could not account for the tension of the compression spring which was a large reason as to the jump height of the hopper. The weight of the dowel used for the base is would have likely taken away from the efficency since it is nearly twice as dense as the hopping mechanism itself.



dy(meters	F(Newtons)
0	0.27468
0.002	11.0853
0.004	13.734
0.006	15.696
0.008	18.1485

Closing Thoughts

In conclusion, I was rather disappointed with my inability to create a working timer for my hopper. I felt that though my hopping mechanism worked so well, more of my time initially should have been spent figuring out a semi functional timer to at the very least have an idea of what I should trying to make. I also wish I would have spent more time branching out. I felt as though once I started working on one aspect, I was stuck to it, and I should have accepted that there were other options and no matter how long I spend on one thing, theres always a better solution. Regardless, I believe that this project allowed me to experiment with my own design process and I now have a better idea how to tackle a project of this nature.